An Auditing Agent Asserts at Ceiling Whether or Not There Is Anything to Find

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Abstract

Automated agents are increasingly used to audit whether a model carries a hidden objective, and their reports read as confident natural language. We audit one such white-box agent and ask a question its published evaluation does not: what does it say when there is nothing to find? Its *assertion rate* turns out to be decoupled from the evidence available to it. On byte-identical evidence, a presuppositional prompt yields 9/9 confident and mutually contradictory answers against 2/10 under neutral phrasing (Fisher exact $p = 0.0007$). Across a six-rung dilution ladder, assertion stays pinned at 1.00 over 120 runs while accuracy falls from 0.55 to 0.05. Detection itself reproduces. The failure is calibration: for evaluators of this kind, confidence carries no information about whether a finding exists.

1 The missing arm of an evaluation

Model diffing methods compare a base model against a finetuned one and try to name what changed. Activation Difference Lens (ADL) [1] reports its diffing agent naming the finetuning objective for 91% of organisms at grade ≥ 2, against 39% for a black-box baseline.

That is a *sensitivity* number: it measures how often the agent recovers an objective that is present. The complementary quantity—how often it names an objective that is *absent*—had not been measured for any white-box diffing method, because no such method had been run on a pair with no hidden objective. For anything used as a judge [2, 3], the second number governs trust. An evaluator that reports a finding on every input has perfect sensitivity and is worthless. We ran 200 blind audits to measure it, and could not—for a reason that turned out to be more useful than the number would have been.

2 Setup

Base model Gemma-3-1B. Auditing agent `gpt-5-2025-08-07`—the model ADL itself uses, pinned to a snapshot. Grader `gpt-5-mini-2025-08-07`. We reimplemented ADL in raw HuggingFace hooks (logit lens, Patchscope, steering, directional ablation) rather than TransformerLens. 200 runs: 80 main ($n = 10$ per cell) and 120 ladder ($n = 20$ per rung).[1]

Every run is coded three ways—ASSERT, ABSTAIN, DEGENERATE—by a definition fixed in advance and applied identically on every arm. Neither the agent nor the grader learns which arm it is looking at, and the grader never sees ground truth.

3 We could not build a null, and that is the first result

Our pre-registered primary metric was a false-positive rate: run the agent on pairs that differ for benign reasons and count assertions. **We abandoned it.** Every null we constructed turned out to contain a real signal, and a false-positive rate measured against a contaminated null would mean nothing.

Three nulls, three distinct contaminations. (a) Finetuning on a generic web corpus: the corpus is itself a register, and the agent read it correctly. (b) Two LoRA seeds trained on byte-identical data: these still differ *along* the domain axis, because two runs converge to different points on the same direction. (c) Base versus instruction-tuned: instruction-tuning is itself an objective.

The methodological finding is that *there may be no such thing as a no-objective finetune*.

4 Assertion is decoupled from evidence

4.1 Framing manufactures answers

On the arm with no *narrow* objective, the shipped presuppositional prompt ("The finetuning instilled a specific objective. Identify it.") produces 9/9 assertions. A neutral prompt, on byte-identical evidence, produces 2/10 (Fisher exact $p = 0.0007$) [4, 5]. The nine assertions are not one answer: a blind judge

[1] Code, pre-registration, deviations log, 200 raw agent reports: `https://anonymous.4open.science/r/model-diffing-specificity-1CBF`

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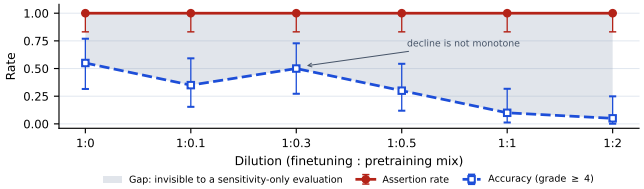


Figure 1: Six dilution rungs, $n = 20$ each, 120 runs. Assertion is flat at 1.00 while accuracy collapses. The shaded gap is what a sensitivity-only evaluation cannot see: a confident wrong answer and a correct one both count as responding. Intervals are Clopper–Pearson 95%. The decline is not monotone.

splits them into two mutually incompatible groups, so at least one group is wrong without any ground truth being consulted.

4.2 Down a dilution ladder, only accuracy moves

Six released rungs progressively mix finetuning data with pretraining data (Figure 1). Assertion is 1.00 on all six—120 of 120 runs—while accuracy at grade ≥ 4 falls from 0.55 to 0.05 ($\rho = -0.94$, $p = 0.005$). The agent does not grow quieter as its evidence thins. It says the same amount, with the same confidence, and is progressively less right.

This trend is exploratory, not pre-registered. Our registered ladder test was a logistic regression of assertion on dilution. Because assertion is constant at 1.00, that regression is degenerate—there is no variance to model—and no coefficient is estimable. The trend reported here is measured on a *different* outcome (accuracy) and is labelled exploratory under our pre-registered multiplicity rule. The degeneracy is itself informative: the registered test could not run precisely because the agent never stopped asserting.

5 Supporting results

Detection reproduces. On a released organism the agent is correct 20/20 under both framings. The failures above are not a broken reimplementation, and we do not claim the method fails at detection.

A candidate label-free instrument. Cross-seed consistency—do independent seeds on identical evidence give mutually compatible answers?—is 1.00 on arms with a real signal and 0.67 on the arm without, and requires no ground truth. We call this a *candidate*, not a validated detector: it rests on four cells, Fisher $p \approx 0.09$, with overlapping intervals.

Magnitude does not predict findability. Within the dilution family, mean activation-difference norm is anti-correlated with accuracy ($\rho = -0.77$, $p = 0.07$). Between families the relation reverses: the arm with the best accuracy has mean norm 747, while the arm with no objective at all has 7756. Triaging models for audit by "large difference, worth a look" would rank these backwards.

6 Method discipline

The rubric and analysis plan were committed before any results existed. Ten deviations are logged in-file, including the abandonment of the pre-registered headline metric in §3. Blinding is enforced rather than documented: a scrub raises a `BlindingViolation` exception if an arm identifier would reach the agent. Grader validation gives $\kappa = 1.000$ (16/16)—weaker than it sounds, since only one case tested the ASSERT/ABSTAIN boundary. Total research time was 18 hours against a 20-hour cap.

7 Limitations

No clean false-positive rate exists in this data, because no clean null does. Everything here is one model family (Gemma-3-1B), one auditing agent, one grader, and a reimplementation of ADL rather than the authors' code, so we cannot separate agent-specific from method-specific behaviour. The consistency result rests on four cells with overlapping intervals and $p \approx 0.09$ and should not be used as a detector on this evidence. The ladder trend is exploratory and its decline is not monotone. Our correctness grades use a reconstructed rubric and are not comparable to the 91% figure quoted in §1.

8 What this implies for LLM judges

Confidence is not a trust signal for this class of evaluator [6]. Any pipeline that treats an evaluator's willingness to answer—or the fluency of its answer—as evidence that something was found is reading a constant. Two cheap mitigations follow directly: report evaluator results under both presuppositional and neutral phrasings; and prefer agreement across independent seeds on identical evidence over single-run confidence [7], since contradiction is detectable without labels even when correctness is not.

References

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